MTH 101

Calculus & Analytical Geometry

Complete Typed Notes

(Lessons 01–23)

Properly Formatted with Tables & Math Mode

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1 Lesson 01: Coordinates, Graphs and Lines

Definition: Calculus is the study of continuous rates of change of quantities.

Examples:

- How distance changes w.r.t time
- How time changes w.r.t speed
- How water flow changes w.r.t time

1.1 Real Numbers

Type	Definition / Examples
Natural Numbers	$1, 2, 3, 4, 5, \ldots$
Whole Numbers	$0, 1, 2, 3, 4, \ldots$
Integers	$\ldots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \ldots$
Rational Numbers	Ratios of integers p/q (division by zero ruled out)
Irrational Numbers	Cannot be expressed as ratio of integers

Note: Why not divide by 0?
If $x \neq 0$, the equation $x/0 = y \Rightarrow x = 0 \cdot y \Rightarrow x = 0$ is contradictory.
If $x = 0$, the equation is satisfied by any number y, so the ratio does not have a unique value. These symbols are undefined.

Historical Note: Pythagoras concluded that the size of a physical quantity must consist of a whole number of units plus some fraction m/n. This idea was shattered in the fifth century B.C. by Hippasus of Metapontum who demonstrated the existence of irrational numbers. He proved that $\sqrt{2}$ is irrational.

Examples of irrational numbers: π, $\cos 19°$, $1 + \sqrt{2}$

1.2 Set Theory

Definition: The collection of well-defined objects is called a set.

Symbol	Meaning
$a \in A$	a is an element of set A
$a \notin A$	a is NOT an element of set A
$\varnothing$ or $\{\}$	Empty set
$A \cup B$	Union of A and B
$A \cap B$	Intersection of A and B
$A \subseteq B$	A is a subset of B

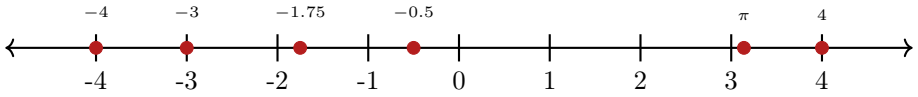
Set-Builder Notation:

$$\{x : x \text{ is a real number and } 2 < x < 3\}$$

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1.3 Coordinate Line

In the 1600's, analytic geometry was developed by the French mathematician Descartes.



One-to-One Correspondence: Each real number corresponds to a single point and each point corresponds to a single real number.

1.4 Order Properties

Definition: For any two real numbers a and b, if $b - a$ is positive, then we say that $b > a$ or that $a < b$.

Property	Rule
(a)	If $a < b$ and $b < c$, then $a < c$
(b)	If $a < b$ and $a + c < b + c$, then $a - c < b - c$
(c)	If $a < b$ and $ac < bc$ when $c > 0$; $ac > bc$ when $c < 0$
(d)	If $a < b$ and $c < d$, then $a + c < b + d$
(e)	If a, b both positive or both negative and $a < b$, then $1/a > 1/b$

1.5 Intervals

Notation	Definition	Type
$[a, b]$	$\{x : a \leq x \leq b\}$	Closed interval
(a, b)	$\{x : a < x < b\}$	Open interval
$[a, b)$	$\{x : a \leq x < b\}$	Half-open
$(a, b]$	$\{x : a < x \leq b\}$	Half-open
$[a, +\infty)$	$\{x : x \geq a\}$	Infinite (closed)
$(-\infty, b]$	$\{x : x \leq b\}$	Infinite (closed)
$(-\infty, +\infty)$	$\mathbb{R}$	Both open and closed

1.6 Solving Inequalities

Example 4: Solve $3 + 7x \leq 2x - 9$

$$\begin{aligned} 7x &\leq 2x - 12 \\ 5x &\leq -12 \\ x &\leq -\frac{12}{5} \end{aligned}$$

Solution set: $\left(-\infty, -\frac{12}{5}\right]$

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Example 5: Solve $7 \leq 2 - 5x < 9$

$$\begin{aligned} 5 &\leq -5x < 7 \\ -1 &\geq x > -\frac{7}{5} \\ -\frac{7}{5} &< x \leq -1 \end{aligned}$$

Solution set: $\left(-\frac{7}{5}, -1\right]$

2 Lesson 02: Absolute Value

Definition:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

Examples:

$$|5| = 5, \quad \left|-\frac{4}{7}\right| = \frac{4}{7}, \quad |0| = 0$$

2.1 Solving Absolute Value Equations

Example 1: Solve $|x - 3| = 4$

$$\begin{aligned} x - 3 = 4 \quad &\text{or} \quad x - 3 = -4 \\ x = 7 \quad &\text{or} \quad x = -1 \end{aligned}$$

Example 2: Solve $|3x - 2| = |5x + 4|$

Case 1:

$$\begin{aligned} 3x - 2 &= 5x + 4 \\ -2x &= 6 \\ x &= -3 \end{aligned}$$

Case 2:

$$\begin{aligned} 3x - 2 &= -(5x + 4) \\ 3x - 2 &= -5x - 4 \\ 8x &= -2 \\ x &= -\frac{1}{4} \end{aligned}$$

Theorem: For any real number a:

$$\sqrt{a^2} = |a|$$

Properties:

Property	Rule
(a)	$\lvert -a \rvert = \lvert a \rvert$
(b)	$\lvert ab \rvert = \lvert a \rvert \lvert b \rvert$
(c)	$\left\lvert \frac{a}{b} \right\rvert = \frac{\lvert a \rvert}{\lvert b \rvert}$

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2.2 Geometric Interpretation

Distance Formula: On a coordinate line:

$$d = |b - a|$$

Expression	Geometric Interpretation
$\|x - a\|$	Distance between x and a
$\|x + a\|$	Distance between x and $-a$
$\|x\|$	Distance between x and the origin

2.3 Inequalities with Absolute Value

Inequality	Equivalent Form
$\|x - a\| < k$	$-k < x - a < k$
$\|x - a\| > k$	$x - a < -k$ or $x - a > k$

Example 1: Solve $|x - 3| < 4$

$$\begin{aligned} -4 < x - 3 < 4 \\ -1 < x < 7 \end{aligned}$$

Interval: $(-1, 7)$

Example 2: Solve $|x + 4| \geq 2$

$$\begin{aligned} x + 4 \leq -2 \quad \text{or} \quad x + 4 \geq 2 \\ x \leq -6 \quad \text{or} \quad x \geq -2 \end{aligned}$$

Set notation: $(-\infty, -6] \cup [-2, +\infty)$

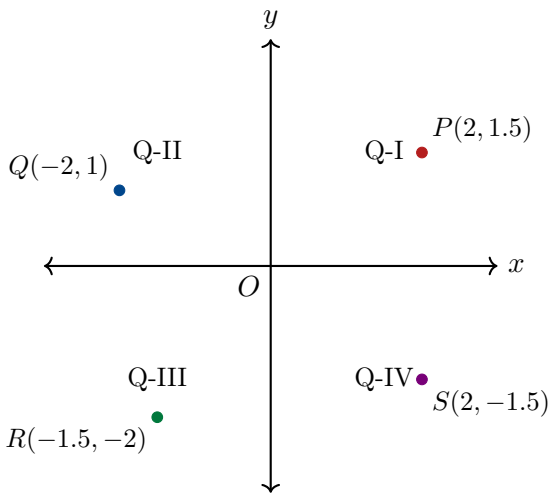
Theorem 1.2.5 (Triangle Inequality): If a and b are any real numbers:

$$|a + b| \leq |a| + |b|$$

3 Lesson 03: Coordinate Planes and Graphs

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3.1 The Coordinate Plane



Quadrants:

Quadrant	Signs
Quadrant I	$x > 0,\ y > 0$
Quadrant II	$x < 0,\ y > 0$
Quadrant III	$x < 0,\ y < 0$
Quadrant IV	$x > 0,\ y < 0$

Definition: The graph of an equation in two variables x and y is the set of all points in the xy-plane whose coordinates are members of the solution set of the equation.

Example: Is $(3, 2)$ a solution of $6x - 4y = 10$?

$$6(3) - 4(2) = 18 - 8 = 10 \quad \checkmark$$

3.2 Intercepts

Example: Find all intercepts of $3x + 2y = 6$

$$x\text{-intercept (set } y = 0\text{): } 3x = 6 \Rightarrow x = 2 \Rightarrow (2, 0)$$
$$y\text{-intercept (set } x = 0\text{): } 2y = 6 \Rightarrow y = 3 \Rightarrow (0, 3)$$

3.3 Symmetry

Type	Test
Symmetric about x-axis	(x, y) and $(x, -y)$
Symmetric about y-axis	(x, y) and $(-x, y)$
Symmetric about origin	(x, y) and $(-x, -y)$

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Example: $y = \frac{1}{8}x^4 - x^2$
Substituting $-x$ for x:

$$y = \frac{1}{8}(-x)^4 - (-x)^2 = \frac{1}{8}x^4 - x^2$$

Equation unchanged $\Rightarrow$ **Symmetric about y-axis**

4 Lesson 04: Lines

Definition 1.4.1 (Slope): If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are points on a non-vertical line:

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

Example: Find the slope through:

Points	Computation	Slope
$(6, 2)$ and $(9, 8)$	$m = \frac{8-2}{9-6} = \frac{6}{3}$	2
$(2, 9)$ and $(4, 3)$	$m = \frac{3-9}{4-2} = \frac{-6}{2}$	-3
$(-2, 7)$ and $(5, 7)$	$m = \frac{7-7}{5-(-2)} = \frac{0}{7}$	0

Theorem 1.4.3: For a non-vertical line:

$$m = \tan\phi$$

If the line is parallel to the y-axis, then $\phi = \pi/2$, so $\tan\phi$ is undefined.

Theorem 1.4.4 (Parallel & Perpendicular):

Condition	Relationship
Parallel	$m_1 = m_2$
Perpendicular	$m_1 m_2 = -1$

Example: Show $A(1, 3)$, $B(3, 7)$, $C(7, 5)$ form a right triangle.

$$m_1 = \frac{7-3}{3-1} = \frac{4}{2} = 2$$
$$m_2 = \frac{5-7}{7-3} = \frac{-2}{4} = -\frac{1}{2}$$
$$m_1 m_2 = (2)\left(-\frac{1}{2}\right) = -1$$

$\Rightarrow AB \perp BC$; thus ABC is a right triangle.

4.1 Equations of Lines

Point-Slope Form:

$$y - y_1 = m(x - x_1)$$

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Slope-Intercept Form:

$$y = mx + b$$

General Form:

$$Ax + By + C = 0$$

Example: Line through $(2, 3)$ with slope $-3/2$

$$y - 3 = -\frac{3}{2}(x - 2)$$
$$y = -\frac{3}{2}x + 6$$

Example: Find slope and y-intercept of $8x + 5y = 20$

$$5y = -8x + 20$$
$$y = -\frac{8}{5}x + 4$$

Slope $m = -\frac{8}{5}$, y-intercept $b = 4$

5 Lesson 05: Distance; Circles, Quadratic Equations

Theorem 1.5.1 (Distance Formula):

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example: Find distance between $(-2, 3)$ and $(1, 7)$

$$d = \sqrt{(1 - (-2))^2 + (7 - 3)^2} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

Theorem 1.5.2 (Midpoint Formula):

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Example: Midpoint of $(3, -4)$ and $(7, 2)$

$$\text{Midpoint} = \left(\frac{3 + 7}{2}, \frac{-4 + 2}{2}\right) = (5, -1)$$

5.1 Circles

Standard Form of a Circle:

$$(x - x_0)^2 + (y - y_0)^2 = r^2$$

Example: Circle of radius 4 centered at $(-5, 3)$

$$(x - (-5))^2 + (y - 3)^2 = 4^2$$
$$(x + 5)^2 + (y - 3)^2 = 16$$
$$x^2 + y^2 + 10x - 6y + 18 = 0$$

Example: Find center and radius of $x^2 + y^2 - 8x + 2y + 8 = 0$

$$(x^2 - 8x + 16) + (y^2 + 2y + 1) = -8 + 16 + 1$$
$$(x - 4)^2 + (y + 1)^2 = 9$$

Center: $(4, -1)$, Radius: 3

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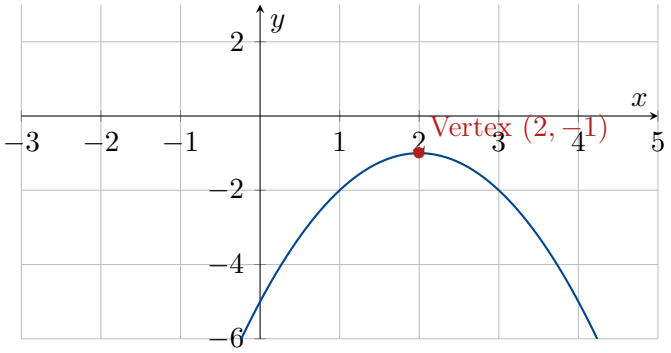
5.2 Quadratic Equations

Standard Form:

$$y = ax^2 + bx + c \quad (a \neq 0)$$

Vertex Formula:

$$x = -\frac{b}{2a}$$



Example: Ball thrown upward: $s = 24.5t - 4.9t^2$

$$\text{Vertex: } t = -\frac{24.5}{2(-4.9)} = 2.5 \text{ sec}$$
$$s = 24.5(2.5) - 4.9(2.5)^2 = 30.625 \text{ m}$$

6 Lesson 06: Functions and Limits

Definition: If a quantity y depends on another quantity x in such a way that each value of x determines exactly one value of y, we say that y is a function of x.

NOT A Function: $y = \pm\sqrt{x}$ — if $x = 4$, then $y = \pm 2$.

Example: $f(x) = x^2$

$$f(3) = 9, \quad f(-2) = 4$$

Example: $\phi(x) = \dfrac{1}{x^3 - 1}$

$$\phi(1) = \frac{1}{1^3 - 1} = \frac{1}{0} = \text{undefined}$$

6.1 Domain of a Function

Example (Natural Domain): $h(x) = \dfrac{1}{(x-1)(x-3)}$

$$\text{Domain: } (-\infty, 1) \cup (1, 3) \cup (3, +\infty)$$

Example (Restricted Domain): $h(x) = \dfrac{x^2 - 4}{x - 2}$

$$h(x) = x + 2, \quad x \neq 2$$

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6.2 Range of a Function

Example: Range of $f(x) = x^2$ is $[0, +\infty)$.

Example: Range of $y = \dfrac{x+1}{x-1}$

$$x = \frac{y+1}{y-1}$$

$y = 1$ is not in the range:

$$\text{Range: } (-\infty, 1) \cup (1, +\infty)$$

7 Lesson 07: Operations on Functions

Operations on Functions:

Operation	Definition
Sum	$(f+g)(x) = f(x) + g(x)$
Difference	$(f-g)(x) = f(x) - g(x)$
Product	$(f \cdot g)(x) = f(x) \cdot g(x)$
Quotient	$(f/g)(x) = f(x)/g(x)$

Composition:

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ consists of all x in the domain of g for which $g(x)$ is in the domain of f.

Example: $f(x) = x^2 + 3$, $g(x) = \sqrt{x}$

$$(f \circ g)(x) = (\sqrt{x})^2 + 3 = x + 3$$

Domain: $[0, +\infty)$

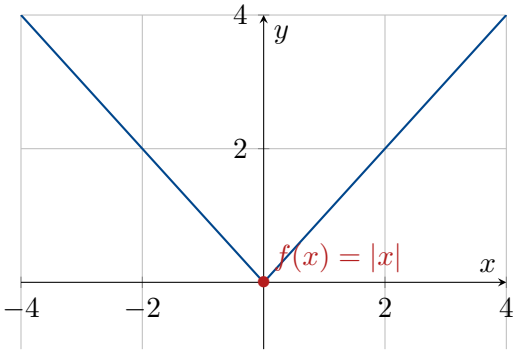
Important: $(f \circ g) \neq (g \circ f)$ generally.

8 Lesson 08: Graphing Functions

Definition: The graph of a function f in the xy-plane is the graph of the equation $y = f(x)$.

Example: Sketch $f(x) = |x|$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$



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Translations:

Transformation	Geometric Effect
$y = f(x) + c$	Translation **UP** by c units
$y = f(x) - c$	Translation **DOWN** by c units
$y = f(x + c)$	Translation **LEFT** by c units
$y = f(x - c)$	Translation **RIGHT** by c units

Vertical Line Test: A graph is the graph of a function if and only if NO vertical line intersects the graph more than once.

9 Lesson 09: Limits (Intuitive Introduction)

Two Branches of Calculus:

- Calculus from the tangent problem → Differential Calculus
- Calculus from the area problem → Integral Calculus

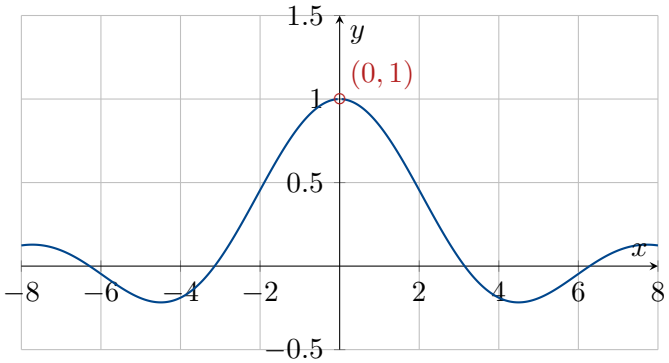
Both depend on the fundamental notion of Limit.

Definition: Limits study the behavior of the y-values of a function in response to the x-values as they approach some number or go to infinity.

Example: $\lim_{x \to 0} \dfrac{\sin(x)}{x}$

$$\lim_{x \to 0^+} \frac{\sin(x)}{x} = 1$$
$$\lim_{x \to 0^-} \frac{\sin(x)}{x} = 1$$

When both match: $\lim_{x \to 0} \dfrac{\sin(x)}{x} = 1$



10 Lesson 10: Limits (Computational Techniques)

Basic Limits:

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Limit	Value
$\lim_{x\to a} k$	k
$\lim_{x\to a} x$	a
$\lim_{x\to+\infty} x$	$+\infty$
$\lim_{x\to-\infty} x$	$-\infty$

Theorem 2.5.1: If $L_1 = \lim f(x)$ and $L_2 = \lim g(x)$ both exist:

$$\lim[f(x)+g(x)] = L_1 + L_2$$
$$\lim[f(x)-g(x)] = L_1 - L_2$$
$$\lim[f(x)\cdot g(x)] = L_1 \cdot L_2$$
$$\lim \frac{f(x)}{g(x)} = \frac{L_1}{L_2} \quad (L_2 \neq 0)$$

Example: $\lim_{x\to 5}(x^2 - 4x + 3)$

$$= (5)^2 - 4(5) + 3 = 25 - 20 + 3 = 8$$

Limits Involving $1/x$:

Limit	Value
$\lim_{x\to 0^+} \frac{1}{x}$	$+\infty$
$\lim_{x\to 0^-} \frac{1}{x}$	$-\infty$
$\lim_{x\to\pm\infty} \frac{1}{x}$	0

Example: $\lim_{x\to 2} \dfrac{x^2-4}{x-2}$

$$= \lim_{x\to 2} \frac{(x+2)(x-2)}{x-2} = \lim_{x\to 2}(x+2) = 4$$

11 Lesson 11: Limits (Rigorous Approach)

Formal Definition of Limit: For any number $\epsilon > 0$, if we can find an open interval (x_0, x_1) containing a such that

$$L - \epsilon < f(x) < L + \epsilon$$

for each x in (x_0, x_1) except possibly $x = a$, then we say

$$\lim_{x\to a} f(x) = L$$

Equivalent Forms:

$$|f(x) - L| < \epsilon \quad \text{and} \quad 0 < |x - a| < \delta$$

Example: Find δ such that $\lim_{x\to 2}(3x - 5) = 1$.

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We need to find $\delta > 0$ such that

$$|(3x-5)-1| < \epsilon \quad \text{whenever} \quad 0 < |x-2| < \delta$$

Step 1: Simplify the expression.

$$\begin{aligned} |(3x-5)-1| &< \epsilon \\ |3x-6| &< \epsilon \\ 3|x-2| &< \epsilon \\ |x-2| &< \frac{\epsilon}{3} \end{aligned}$$

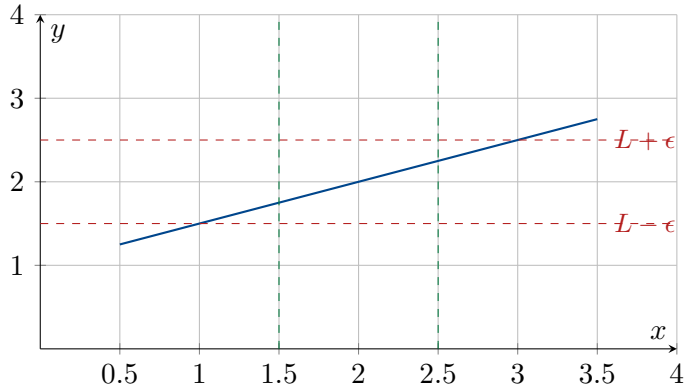
Step 2: Choose δ.

$$\boxed{\delta = \frac{\epsilon}{3}}$$

Step 3: Verify. If $0 < |x-2| < \delta = \epsilon/3$, then

$$|(3x-5)-1| = 3|x-2| < 3 \cdot \frac{\epsilon}{3} = \epsilon$$

Hence $\lim_{x\to 2}(3x-5) = 1$. ■



12 Lesson 12: Continuity

Definition 2.7.1 (Continuity): A function f is continuous at $x = c$ if:

(a) $f(c)$ is defined

(b) $\lim_{x\to c} f(x)$ exists

(c) $\lim_{x\to c} f(x) = f(c)$

If any condition fails, f is **discontinuous** at c.

Example: $g(x) = \begin{cases} \dfrac{x^2-4}{x-2} & x \neq 2 \\ 3 & x = 2 \end{cases}$

$$\lim_{x\to 2} g(x) = \lim_{x\to 2}(x+2) = 4 \neq g(2) = 3$$

So g is discontinuous at $x = 2$.

Theorem 2.7.2: Polynomials are continuous functions.

Theorem 2.7.3 (Properties): If f and g are continuous at c, then:

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(a) $f + g$ is continuous at c

(b) $f - g$ is continuous at c

(c) $f \cdot g$ is continuous at c

(d) f/g is continuous at c if $g(c) \neq 0$

Theorem 2.7.5 (Composition): If $\lim g(x) = L$ and f is continuous at L:

$$\lim f(g(x)) = f(L)$$

Theorem 2.7.9 (Intermediate Value Theorem): If f is continuous on $[a, b]$ and C is any number between $f(a)$ and $f(b)$ inclusive, then there is at least one number x in $[a, b]$ such that $f(x) = C$.

Theorem 2.7.10: If f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, then there is at least one solution of $f(x) = 0$ in (a, b).

13 Lesson 13: Limits and Continuity of Trigonometric Functions

Theorem 2.8.1: The functions $\sin(x)$ and $\cos(x)$ are continuous.

$$\lim_{x\to 0} \sin(x) = 0, \quad \lim_{x\to 0} \cos(x) = 1$$

Proof (for sin):

$$\begin{aligned}\lim_{h\to 0} \sin(c+h) &= \lim_{h\to 0} [\sin(c)\cos(h) + \cos(c)\sin(h)] \\ &= \sin(c) \lim_{h\to 0} \cos(h) + \cos(c) \lim_{h\to 0} \sin(h) \\ &= \sin(c)(1) + \cos(c)(0) = \sin(c)\end{aligned}$$

Continuity of Other Trig Functions:

Function	Continuous except where
$\tan(x) = \dfrac{\sin(x)}{\cos(x)}$	$\cos(x) = 0$
$\cot(x) = \dfrac{\cos(x)}{\sin(x)}$	$\sin(x) = 0$
$\sec(x) = \dfrac{1}{\cos(x)}$	$\cos(x) = 0$
$\csc(x) = \dfrac{1}{\sin(x)}$	$\sin(x) = 0$

Squeeze Theorem: Let $g(x) \leq f(x) \leq h(x)$ for all x in some open interval containing a. If

$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L$$

then

$$\lim_{x\to a} f(x) = L$$

Example: $\lim_{x\to 0} x^2 \sin^2\left(\dfrac{1}{x}\right)$

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Since $0 \leq \sin^2(1/x) \leq 1$:

$$0 \leq x^2 \sin^2(1/x) \leq x^2$$
$$\lim_{x\to 0} 0 = \lim_{x\to 0} x^2 = 0$$

So by Squeeze Theorem:

$$\boxed{\lim_{x\to 0} x^2 \sin^2(1/x) = 0}$$

Theorem 2.8.3 (Important Trig Limits):

$$\lim_{x\to 0} \frac{\sin(x)}{x} = 1$$
$$\lim_{x\to 0} \frac{1-\cos(x)}{x} = 0$$

14 Lesson 14: Tangent Lines, Rates of Change

Slope of Secant Line:

$$m_{\text{sec}} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Slope of Tangent Line:

$$m_{\tan} = \lim_{x_1\to x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Average Velocity:

$$\text{Average Velocity} = \frac{f(t_1) - f(t_0)}{t_1 - t_0}$$

Definition 3.1.1 (Average Rate of Change): If $y = f(x)$, the average rate of change of y w.r.t x over $[x_0, x_1]$ is

$$m_{\text{sec}} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Definition 3.1.1 (Instantaneous Rate of Change):

$$m_{\tan} = \lim_{x_1\to x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Example: $y = f(x) = x^2 + 1$

(a) Average rate over $[3, 5]$:

$$m_{\text{sec}} = \frac{f(5) - f(3)}{5 - 3} = \frac{26 - 10}{2} = 8$$

(b) Instantaneous rate at $x_0 = -4$:

$$\begin{aligned} m_{\tan} &= \lim_{x_1\to -4} \frac{(x_1^2 + 1) - 17}{x_1 + 4} \\ &= \lim_{x_1\to -4} \frac{x_1^2 - 16}{x_1 + 4} \\ &= \lim_{x_1\to -4} (x_1 - 4) = -8 \end{aligned}$$

(c) Instantaneous rate at general point:

$$m_{\tan} = \lim_{x_1\to x_0} \frac{x_1^2 - x_0^2}{x_1 - x_0} = 2x_0$$

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